LEVEL7/YEAR3,RP NGOMA

PYTHON and FUNDAMENTALS OF AI

GROUP PRACTICAL ASSIGNMENT AND PRESENTATION(FORM 6 GROUPS)

Facilitator: Eng. Leopord UWAMAHORO

Group No.	_____	**Class**	_____
Group Leader	_____	**Date**	_____
Presentation Date	_____8.09.2026_____	**Duration**	_____
Number of Members	_____	**Submission**	Digital folder + presentation

1. Purpose of the Assignment

This assignment is designed to make students prepare and use a working Python environment, apply fundamental Python programming concepts, use Python libraries, develop and test programs, and communicate their work through a group presentation. The practical work follows the handout's progression from preparing the Python environment to developing Python concepts and scripts.

2. Important Instructions

- This is group work for development and presentation; however, every student must participate actively.
- EVERY STUDENT MUST COME WITH A COMPUTER and must be able to run Python on that computer.
- Before starting the programming tasks, each student must configure the Python environment and install/verify the required libraries.
- The group must not rely on only one member's computer. During assessment, any member may be asked to demonstrate a task on his/her computer.
- All members must understand the submitted code and be ready to explain or modify part of it during the presentation.
- Keep evidence of environment configuration, commands, code execution, outputs, errors corrected, and final results.
- Use the supplied AI with Python handout as the main learning reference for the assignment.

3. Part A - Individual Computer and Python Environment Preparation

Before the group starts the main programming work, EVERY MEMBER must complete the following on his/her own computer.

A1. Verify/Install Python

1. Check whether Python 3 is installed. If it is not installed, install it.
2. Verify the installed Python version from Command Prompt/Terminal.
3. Ensure Python can be executed from the command line and correct the PATH if necessary.

4. Keep a screenshot showing successful Python version verification.

A2. Configure a Development Tool

5. Install or configure a suitable Python editor/IDE such as Visual Studio Code, PyCharm, Jupyter Notebook/JupyterLab, or another appropriate environment.
6. Create a working folder named AI_With_Python_Assignment.

A3. Verify PIP and Install Libraries

7. Verify that PIP is available and display its version.
8. Use PIP to install the libraries required by your group's work.
9. At minimum, for the data/AI activities prepare NumPy, pandas, Matplotlib, and scikit-learn.
10. Create environment_test.py, import the installed libraries, run it, and confirm that no import error occurs.
11. Keep screenshots of the important commands and successful execution.

4. Part B - Python Fundamentals Practical

Develop a small Student Performance Program. It must request a student's name and marks for at least three subjects, store the values using appropriate variables/data types, calculate total and average, and display well-formatted results.

- Use input and output statements.
- Demonstrate variables and appropriate data types.
- Use arithmetic, comparison and logical operators where appropriate.
- Use conditional statements to classify performance (for example: Excellent, Very Good, Good, Pass or Fail).
- Reject or handle invalid marks such as values below 0 or above 100.
- Use comments to make important parts of the code understandable.

5. Part C - Collections and Repetition

Extend the program to process several students and demonstrate the collection concepts covered in the handout.

- Use a loop to process more than one student.
- Use an appropriate list, tuple, set and/or dictionary in meaningful parts of the solution.
- Display all stored student records.
- Determine the highest and lowest average.
- Calculate the overall class average.
- Count students who pass and fail.
- Be prepared to explain why each selected collection is appropriate.

6. Part D - File Handling and Exception Handling

- Save processed student results to a file.
- Read the saved results back and display them.

- Use appropriate file modes for reading/writing.
- Use try/except where appropriate so that common input or file errors do not simply crash the program.
- Document at least one error encountered during development and explain how the group corrected it.

7. Part E - Python Libraries, Data Analysis and Visualization

Using the results produced by the program or an instructor-approved dataset:

- Load and organize data using pandas and/or NumPy.
- Calculate useful summary information.
- Produce at least one clear Matplotlib graph comparing student performance.
- Add a meaningful title and axis labels to the graph.
- Write a short interpretation of what the graph shows.

8. Part F - Introductory AI/Machine Learning Practical

Using an instructor-approved dataset, demonstrate a simple machine-learning workflow with scikit-learn. The purpose is to show that the group can connect Python programming and libraries to an introductory AI task.

12. Load the dataset.
13. Identify the input feature(s) and target/output.
14. Check the data and prepare it as required.
15. Separate features from the target.
16. Split the data into training and testing sets.
17. Select and train one appropriate simple machine-learning model.
18. Generate predictions and evaluate them using an appropriate metric.
19. Explain, in simple terms, what the obtained result means.

9. Group Presentation

Each group will present and demonstrate its work. The presentation must be practical: slides may support the explanation, but the group must also run the programs and show evidence from members' computers.

Presentation Section	Expected Content
1. Introduction	Problem/task, group members and what the group developed.
2. Environment Preparation	Python version, IDE, PATH/PIP configuration and installed libraries.
3. Python Concepts Used	Variables, data types, operators, conditions, loops and collections.
4. Program Demonstration	Run the student performance program and explain the main logic.
5. File Handling & Errors	Show saved/read data and explain one handled error.

6. Data Analysis & Graph	Show the analysis and interpret the graph.
7. AI/ML Demonstration	Explain features/target, training/testing, model, prediction and evaluation.
8. Challenges & Conclusion	What was difficult, how it was solved, and what the group learned.

10. Participation Requirement

IMPORTANT: The presentation is a GROUP presentation, but marks should reflect individual participation. Every member must present or demonstrate a meaningful part. The facilitator may select any member to explain code, run the program, identify a library, explain an error, or make a small modification.

11. What the Group Must Submit

- One organized group assignment folder.
- Python source files (.py) and/or notebook files (.ipynb).
- Dataset(s) used, where applicable.
- Generated result/data files.
- Graph(s) produced by the program.
- requirements.txt listing required external libraries.
- Presentation slides.
- Evidence/screenshots showing that group members prepared their Python environments.
- A short README file identifying group members and explaining how to run the project.

12. Assessment Rubric (100 Marks)

Assessment Area	**Marks**
Individual environment preparation and verification	15
Python fundamentals and program correctness	15
Conditions, loops and collections	15
File handling, testing and exception handling	10
Use of Python libraries and data analysis	10
Visualization and interpretation	10
Introductory AI/ML implementation and explanation	10
Group presentation and live demonstration	10
Individual participation and ability to explain work	5
TOTAL	**100**

13. Preparation Checklist for Every Student

☐ I have a computer available for the practical/presentation.

☐ Python 3 runs successfully on my computer.

☐ I can check the Python version from the terminal.

☐ PIP works on my computer.

☐ I installed/verified the libraries needed by the group.

☐ I can open and run the group's Python program.

☐ I understand the section of code I contributed.

☐ I can explain the program's inputs, processing and outputs.

☐ I am ready to demonstrate my contribution during the presentation.

END OF ASSIGNMENT